Course Code	Course Title	Credit
CSC801	**Advanced Artificial Intelligence**	**3**

Prerequisite: Engineering Mathematics, Data Structures and Algorithm, Python Programming			
Course Objectives:			
1	To relate with the basic concepts of Probabilistic Models.		
2	To understand the scope of Generative Networks in the field of AI.		
3	To recognize various components of Autoencoder Architecture and Training process.		
4	To learn the fundamentals of Transfer Learning.		
5	Provide students with a comprehensive understanding of ensemble methods and their applications.		
6	To explore the nascent applications of AI		
Course Outcomes: After successful completion of the course student will be able to			
1	Acquire basic knowledge of Probabilistic Models.		
2	Analyze the working and architecture for Generative Networks.		
3	Interpret various components and various types of Autoencoders		
4	Understand various aspects of Transfer Learning.		
5	Apply ensemble learning techniques to real-world problems and demonstrate improved predictive performance.		
6	Relate to the nascent technologies in the field of artificial intelligence.		
Module		**Content**	**Hrs**
1.0		**Generative and Probabilistic Models**	**08**
	1.1	**Introduction:** Overview of generative models and their importance in AI, Fundamentals of Probability theory and generative modeling, Introduction to GANs, VAEs and other generative models. Significance of generative models, Challenges with generative models.	
	1.2	**Probabilistic Models:** Gaussian Mixture Models (GMMs), Hidden Markov Models (HMMs), Bayesian Networks, Markov Random Field (MRFs), Probabilistic Graphical Model.	
2.0		**Generative Adversarial Network**	**07**
	2.1	**Basics of GAN :** Generative Adversarial Networks (GANs) architecture, The discriminator model and generator model, Architecture and Training of GANs, Vanilla GAN Architecture. GAN variants and improvements (DCGAN,WGAN, Conditional GAN, CycleGAN), Challenges- Training instability and model collapse, GAN applications in image synthesis and style transfer.	
3.0		**Variational Autoencoders**	**07**

	3.1	**Introduction:** Basic components of Variational Autoencoders(VAEs), Architecture and training of VAEs the loss function, Latent space representation and inference, Applications of VAEs in image generation. **Types of Autoencoders:** Undercomplete autoencoders, Sparse autoencoders, Contractive autoencoders, Denoising autoencoders, Variational Autoencoders (for generative modelling)	
4.0		**Transfer Learning**	**05**
	4.1	**Introduction to transfer learning** Basic terminologies, Pre-trained model and data sets, Feature extraction and fine tune transfer learning , Recent advancement in transfer learning : self-supervised learning and meta learning.	
5.0		**Ensemble learning**	**06**
	5.1	**Ensemble Classifiers**: Introduction to Ensemble Methods. Bagging and random forests, Boosting algorithms: AdaBoost Stacking and Blending Models, Extreme Gradient Boosting (XGBoost): XGBoost Regression and classification.	
6.0		**Nascent Technologies in AI**	**06**
	6.1	**Ensemble Classifiers**: Limitations of 2D Learning Environments, Evolution of virtual worlds and immersive technologies, Definition and concepts of Augmented Reality, Definition and concept of the Metaverse, Characteristics and components of the Metaverse, Challenges and opportunities in the Metaverse ecosystem, AI in the realm of emerging quantum paradigms.	

Textbooks:	
1	Foster, D., 2022. *Generative deep learning*. " O'Reilly Media, Inc.".
2	Koller, D. and Friedman, N., 2009. *Probabilistic graphical models: principles and techniques*. MIT press
3	Goodfellow, I., 2016. Deep Learning-Ian Goodfellow, Yoshua Bengio, Aaron Courville- Google Books
4	Murphy, K.P., 2012. *Machine learning: a probabilistic perspective*. MIT press
5	Zhou, Z.H., 2012. *Ensemble methods: foundations and algorithms*. CRC press.

References:	
1	Xiong, J., Hsiang, E.L., He, Z., Zhan, T. and Wu, S.T., 2021. Augmented reality and virtual reality displays: emerging technologies and future perspectives. *Light: Science & Applications*, *10*(1), p.216.
2	Mystakidis, S., 2022. Metaverse. *Encyclopedia*, *2*(1), pp.486-497
3	Gill, S.S., Xu, M., Ottaviani, C., Patros, P., Bahsoon, R., Shaghaghi, A., Golec, M., Stankovski, V., Wu, H., Abraham, A. and Singh, M., 2022. AI for next generation computing: Emerging trends and future directions. *Internet of Things*, *19*, p.100514